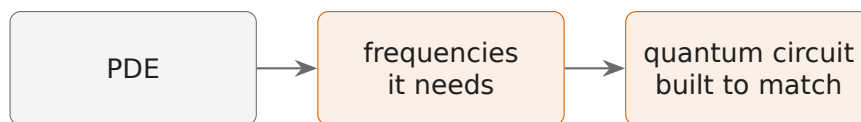


Qapiino

Designing quantum circuits from
the physics they have to solve

Spectrum-Matched Quantum-Assisted Physics-Informed Neural Networks



WISER Summer Program 2026 • Industry Challenge (BQP) • Top 10

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About the author

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Role. Designed the SMCD method, built the simulator, training and explainability pipeline, ran and adjudicated the pre-registered experiments, and wrote this report.

Abstract

Physics-informed neural networks learn smooth, slow variations quickly and sharp, fast ones slowly. A quantum circuit that re-uploads its input can only produce a fixed, predictable set of frequencies. We turned that into a design rule, SMCD: read the frequencies a PDE's solution needs from the equation, then build the circuit whose frequency set covers them, with no architecture search. Our model mixes eight copies of the designed circuit through a linear output layer.

We applied it to a real planning problem: waterlogging of farmland between two rivers, fed by six leaking irrigation canals. Predictions were committed to git before training, the model was tested on seeds never used in development, and a script issued every verdict. The model met all five predictions. It reproduces the water table to 0.09% error, 18× more accurately than a classical network of the same size and 9.9× more accurately than the same circuit with random frequencies, and it places the waterlogged stretch of the field to within 1 m (714 m vs. 713 m exact). It also answers the planner's question directly: lining the canals to cut seepage by 20% dries the whole field. On the Poisson benchmark the designed circuit beats its random-frequency twin by 236× and a Fourier-feature network by three orders of magnitude.

At a glance

0.09%

error on the farmland
water table, held-out seeds

18×

more accurate than
a classical network
of the same size

5/5

pre-registered
predictions met

1 m

error in the waterlogged
length (714 m vs. 713 m)

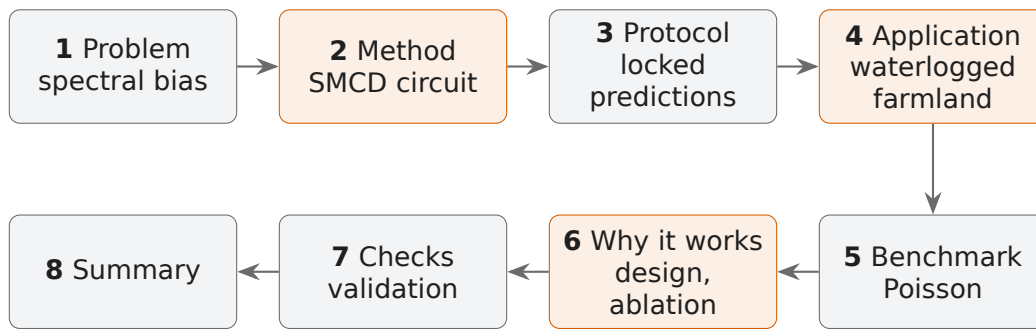
236×

designed vs. ran-
dom frequencies,
Poisson benchmark

20%

less canal seepage (lining)
dries the whole field

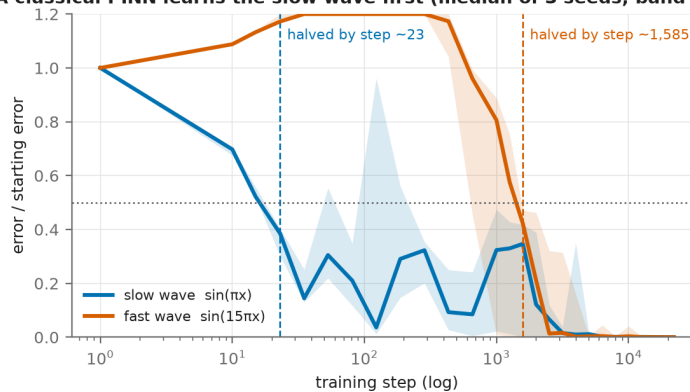
How this report flows



1 The problem: neural networks learn slow waves first

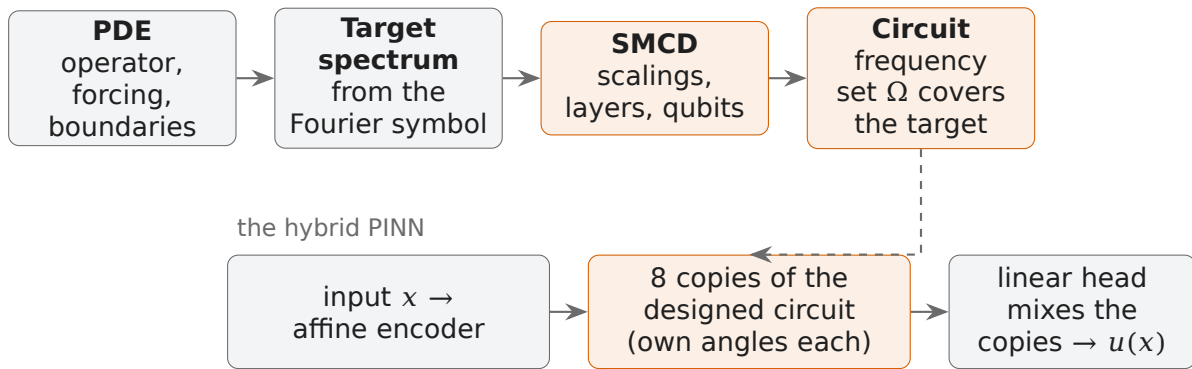
A standard physics-informed neural network (PINN) fits a smooth solution quickly and struggles with fine detail. This is *spectral bias*: the network's learning speed falls with frequency. Below, a classical network halves its error on the slow $\sin \pi x$ part of a Poisson solution within 23 steps, and needs 658–1,585 steps for the fast $\sin 15\pi x$ part.

A classical PINN learns the slow wave first (median of 3 seeds, band = range)

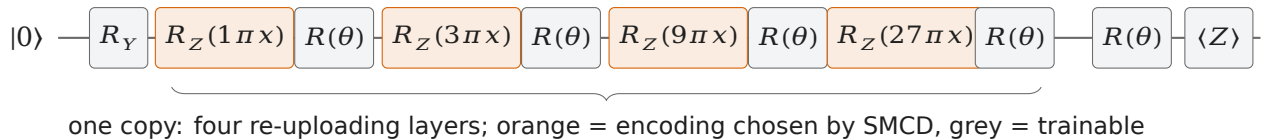


A re-uploading quantum circuit behaves differently. Its output is a Fourier series whose frequencies are fixed by how the input is encoded, so they can be chosen in advance.

2 Method: build the circuit from the equation



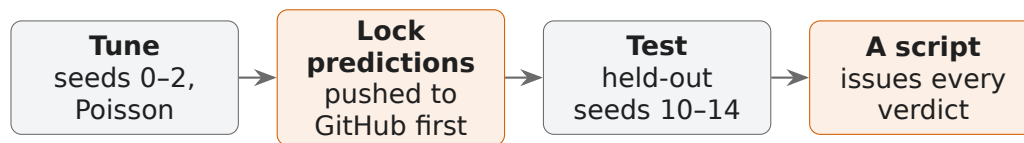
Example. Poisson's solution $\sin \pi x + 0.3 \sin 15\pi x$ needs the frequencies π and 15π . SMCD picks encoding scalings $\pi \cdot \{1, 3, 9, 27\}$ on one qubit with four layers; the circuit can then produce every multiple of π up to 40π , which covers both. No search, and the design is checkable before training.



Encoding scalings	$\pi \cdot \{1, 3, 9, 27\}$
Reachable frequencies Ω	every sum $\pm\pi \pm 3\pi \pm 9\pi \pm 27\pi =$ all multiples of π from -40π to 40π (81)
Needed by the solution	π and 15π (both in Ω)
Copies	8, each a one-qubit circuit with its own angles, same Ω
Trainable parameters	91

Why eight copies: in a single circuit every wave amplitude depends on the same few angles, so improving one frequency disturbs another. Separate copies mixed by a linear head give the model an independent handle on each amplitude (Section 6).

3 How we tested it

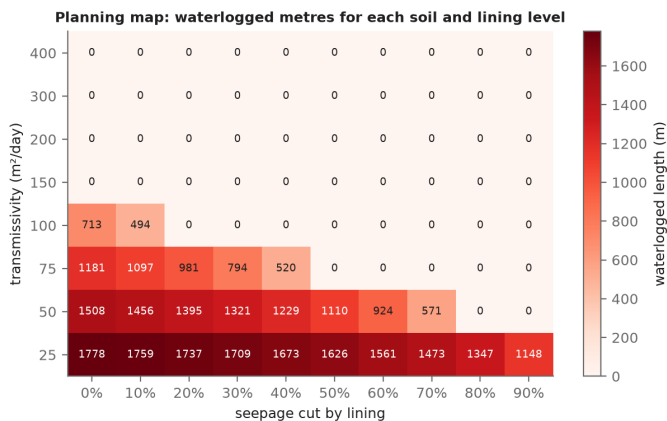
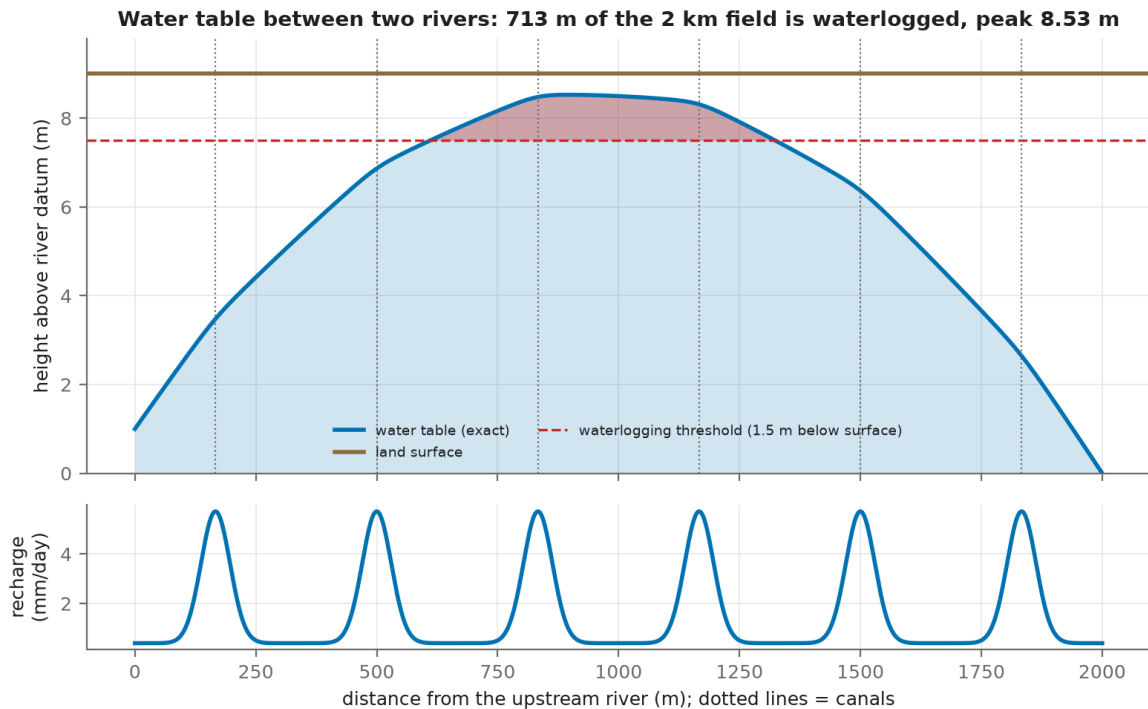


Problems	waterlogged farmland (groundwater flow), Poisson benchmark
Our model	SMCD circuit, 8 copies (91 parameters)
Baselines	the same circuit with random frequencies (91); an MLP of the same size (91); RFF matched, a classical model given the same frequencies (83); Fourier features (129); a large MLP (8,513)
Training	20,000 Adam + 2,000 L-BFGS steps, exact boundary conditions, every model at the learning rate tuning chose for it
Rule	a prediction is confirmed only if its pre-set threshold is met and, for comparisons, all five seeds agree

Tuning numbers are optimistic by construction, so no claim in this report rests on them. Every result below comes from seeds 10-14, which played no part in choosing the model, scored against predictions that were on GitHub before those runs started.

4 Application: waterlogged farmland

Two rivers 2 km apart bound an aquifer under farmland irrigated by six leaking canals. Where the water table rises within 1.5 m of the surface, crop roots drown. The canal spacing tells SMCD the frequencies the solution needs before any training.



What the model tells a planner

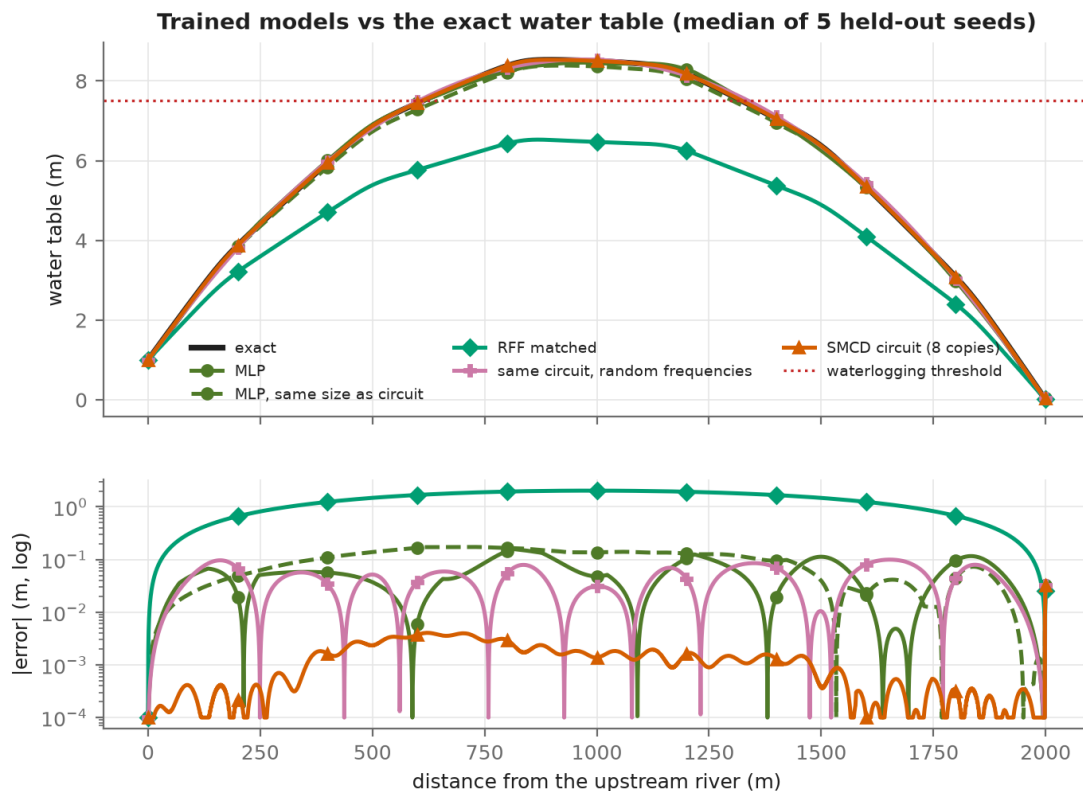
- 713 m of the field is waterlogged to-day (peak 8.53 m).
- Lining the canals to cut seepage by 10% leaves 494 m; by 20%, none.
- On clay loam the same canals water-log 1778 m.

Can the trained model answer it?

We locked five predictions in git before training on the held-out seeds, then scored them with a script.

Prediction	SMCD circuit
G-1 error below 1%	yes (0.09%)
G-2 beats a same-size classical network	yes (18×)
G-3 beats the same circuit with random frequencies	yes (9.9×)
G-4 improvement lies on the designed frequencies	yes (99.4%)
G-5 waterlogged length within 5% of the truth	yes (714 m vs 713 m)
Met	5 of 5

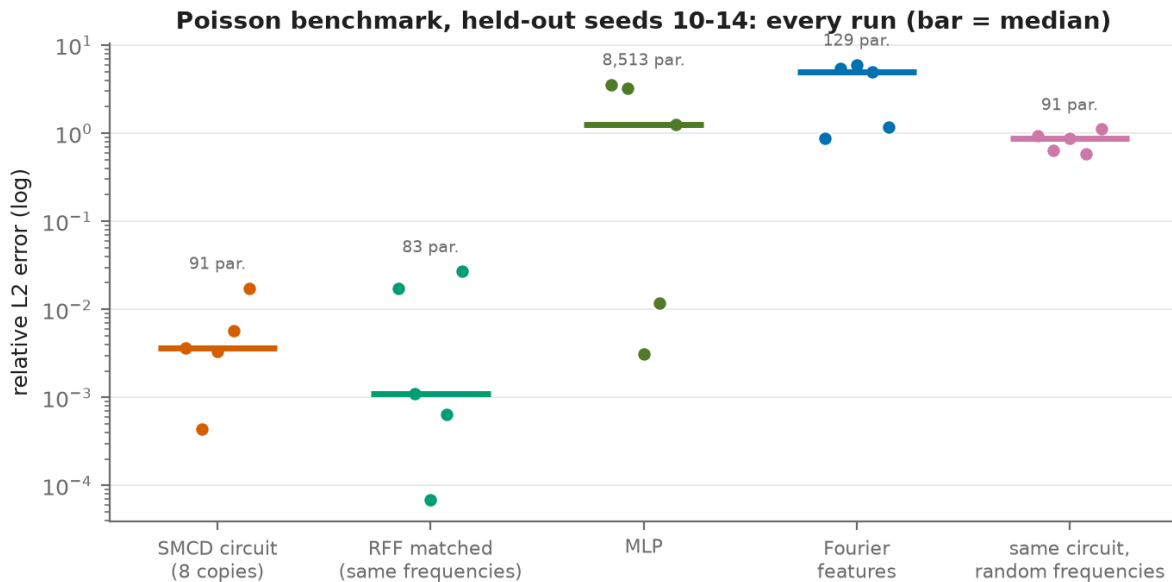
Model	parameters	median error
SMCD circuit	91	0.09%
same circuit, random frequencies	91	0.92%
MLP, same size	91	1.7%
MLP, large	8,513	1.9%
RFF matched (same frequencies)	83	23%



The circuit tracks the true water table to within a few millimetres across the whole field and gets the waterlogged length right to 1 m. It is also more accurate than an MLP with 94 times as many parameters.

5 Benchmark: Poisson

The Poisson benchmark, with solution $\sin \pi x + 0.3 \sin 15 \pi x$ mixes a slow and a fast wave, the case spectral bias makes hard. Eight predictions were locked before the held-out runs; **5 of 8 held**.



- **Held:** beats the same circuit with random frequencies (236×) and Fourier features (1,360×); its improvement lands on the designed frequencies (99.9%); the NTK is flatter inside Ω than outside; the encoder stays put during training.
- **Did not hold:** a classical model given the same frequencies (83 parameters) is 3× more accurate here; the circuit does not leave the lazy training regime earlier.
- **Inconclusive:** against the large MLP, which failed on three of five held-out seeds and did well on two.

Every one of the circuit's runs landed below 2% error, while the MLP and Fourier features were unstable at the learning rates tuning chose for them.

6 Why it works

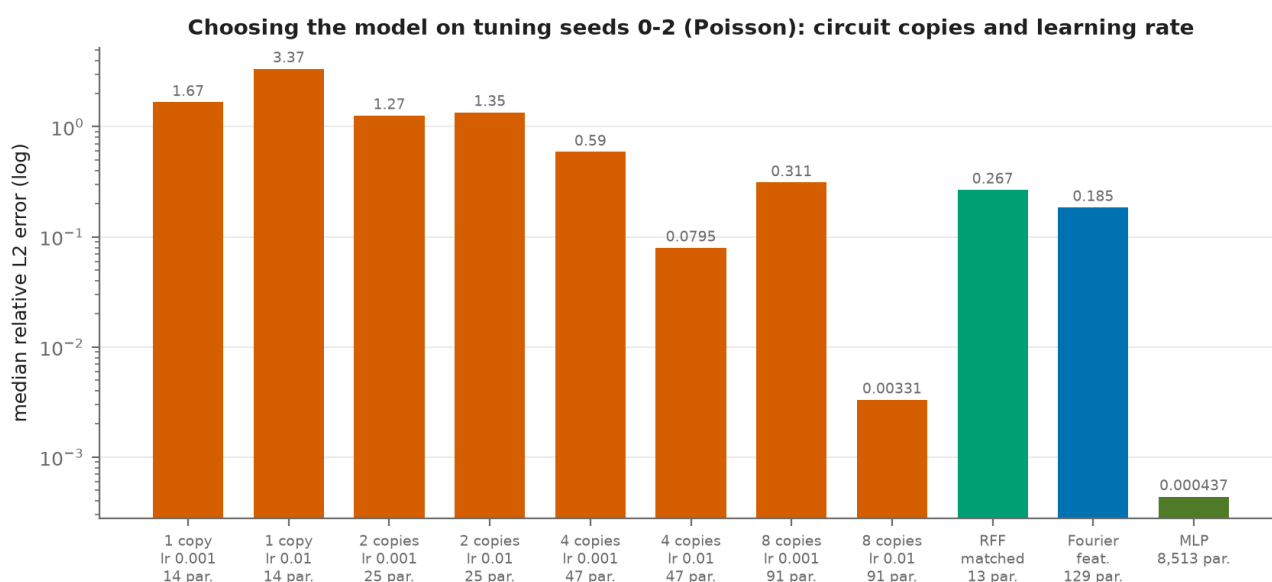
The designed frequencies matter

Same circuit, same size, same training; only the frequencies differ. With the designed frequencies the error is 9.9× lower on the farmland problem and 236× lower on Poisson, on every held-out seed. The effect already shows in a single copy: on the heat equation the designed circuit beats its random twin on every seed (95% interval 2.5–17.7×).

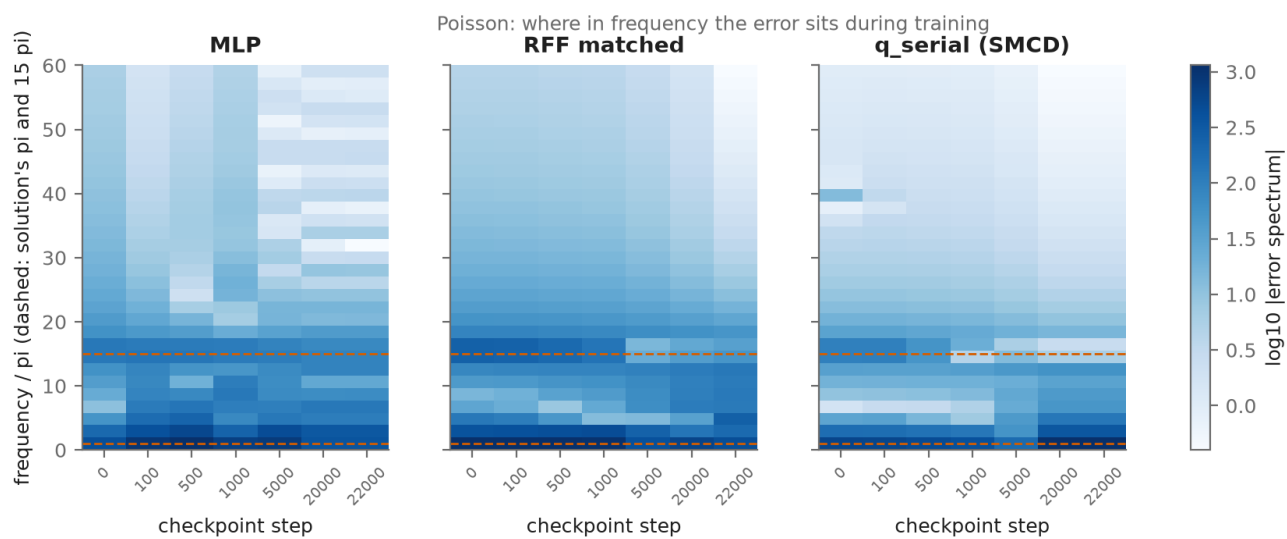


Eight copies, not one

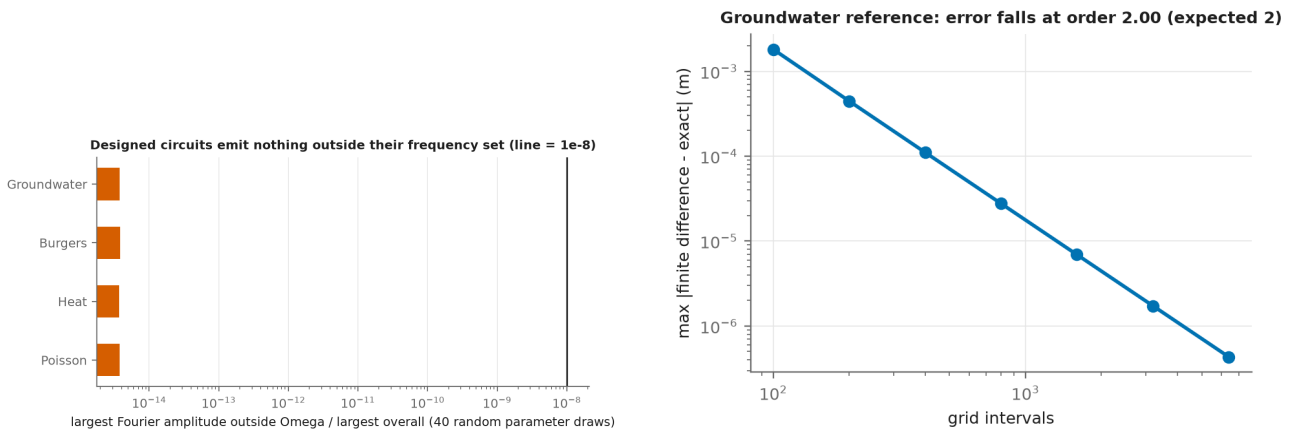
We chose the number of copies and the learning rate on the tuning seeds. One copy (14 parameters) cannot fit Poisson at either learning rate; eight copies reach 0.3%. The same holds on held-out data: on the farmland problem a single copy scores 53% error against 0.09% for eight.



The per-frequency error shows the mechanism in a single copy. It removes the hard 15π error early, then loses the easy π component: every amplitude is tied to the same ten angles, and the loss weights the 15π mode 225 times more, so fixing one costs the other. Eight copies with a linear head set the amplitudes independently.



7 Checks



Left: the circuit builds exactly what SMCD designs. At random parameters, every designed circuit's output outside Ω is at most 4×10^{-15} of its largest component. Right: our reference solver converges at the textbook rate, so the errors we report are the models', not the reference's. The farmland problem's exact solution also matches an independent finite-difference solver to 5×10^{-8} m.

Summary

- **A design rule that works as specified.** SMCD reads the needed frequencies from the equation and builds a circuit that produces exactly those, with no search.
- **Accurate on a real problem.** On waterlogged farmland the model reaches 0.09% error, 18× better than a classical network of the same size, and gets the waterlogged length right to 1 m.
- **The design is the reason.** The same circuit with random frequencies is 9.9× worse on farmland and 236× worse on Poisson.
- **Scored strictly.** Predictions locked on GitHub before the runs, held-out seeds, and a script issuing every verdict: 5 of 5 on farmland.
- **Next.** Run the circuits on quantum hardware (everything here is simulated), and close the gap to the frequency-matched classical model on Poisson.

Links

Code, data, every run and every figure: https://github.com/Mannava-Daasaradhi/QAPINN_Spectra (branch v2). The full technical paper, [paper/main.pdf](#) in the repository, records every experiment, including the model configurations we did not adopt.